From Annotation Agreement to Dialogue Coherence: A Grice-Based Evaluation of LLM Re-Annotations on MultiWOZ

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This paper examines whether annotation mismatch can function as an indirect signal of dialogue incoherence in the MultiWOZ dialogue corpus. To address this question, we combine two analytical branches. In the first branch, we re-annotate MultiWOZ with an LLM prompt and compare the resulting symbolic annotations to the original annotation layer using BLEU and Jaccard under strict, normalized, and lenient matching regimes. In the second branch, we evaluate the raw dialogue turns with a Grice-inspired, role-aware coherence judge that scores each next turn in context according to Quality, Quantity, Relation, and Manner. The resulting coherence judgments are then aggregated and related to annotation overlap at both dialogue and turn level. The results show that the relationship between annotation mismatch and coherence is neither uniform nor stable across levels of analysis. At the dialogue level, lower overlap does not meaningfully correspond to lower average coherence, with BLEU correlations remaining near zero ($\rho = -0.010$ to -0.023). At the turn level, the pattern is more differentiated: BLEU-1 shows a small but consistent positive relationship with coherence ($\rho = 0.142$-0.167), whereas Jaccard and BLEU-4 do not show the same stable behavior. The study therefore suggests that annotation agreement and pragmatic coherence should not be treated as interchangeable indicators of dialogue quality. More broadly, it argues for evaluation designs that distinguish between symbolic agreement and discourse-level quality while relating them analytically.

CCS CONCEPTS • Discourse, dialogue and pragmatics • Natural language processing • Empirical studies in HCI

Additional Keywords and Phrases: MultiWOZ, task-oriented dialogue, large language models, annotation agreement, dialogue coherence, BLEU, Jaccard, Gricean coherence

1 INTRODUCTION

Task-oriented dialogue research depends heavily on annotated benchmark corpora, and MultiWOZ remains one of the most widely used resources in this area. At the same time, the history of MultiWOZ makes clear that annotation quality is not a trivial background issue but a central methodological concern. MultiWOZ 2.4 was introduced specifically to correct substantial annotation problems in earlier versions of the dataset, with corrections affecting more than 40% of turns and more than 65% of dialogues [1]. This raises a broader question that goes beyond dataset maintenance: if annotation layers are themselves historically noisy and revisable, what exactly does agreement with an existing annotation scheme tell us about the underlying dialogue?

A second development sharpens this question further. Large language models are increasingly used both as annotators and as evaluators. Recent work suggests that LLMs can produce useful annotations efficiently, but also that their outputs are sensitive to prompt design, task framing, and the subjective character of the annotation problem [2], [3], [4]. In parallel, LLMs are increasingly being explored as automatic dialogue judges, especially for coherence- and quality-related assessment [5], [6]. However, these two lines of work are usually treated separately: one asks whether LLM outputs align with human or legacy labels, while the other asks whether a dialogue turn or dialogue as a whole is coherent, appropriate, or interactionally successful.

This paper brings these strands together by asking whether annotation mismatch can function as an indirect signal of dialogue incoherence. More specifically, we investigate whether lower agreement between our LLM-generated MultiWOZ-style re-annotations and the original MultiWOZ annotation layer tends to occur in dialogues or turns that are themselves pragmatically noisier, more confusing, or less coherent. The basic intuition is that locally clear and interactionally well-formed turns should be easier to annotate consistently, whereas pragmatically weak or ambiguous turns may invite greater divergence between two annotation versions. Rather than treating annotation overlap as a quality measure in its own right, we therefore examine whether it is meaningfully related to a second, explicitly discourse-oriented notion of quality.

To do so, we build a two-branch pipeline. In the first branch, we re-annotate the MultiWOZ 2.3 Corpus with an LLM prompt and compare the resulting symbolic annotations to the original corpus annotations using BLEU and Jaccard under strict, normalized, and lenient matching regimes. In the second branch, we evaluate the raw dialogue turns with a Grice-inspired coherence judge that scores each next turn in context according to Quality, Quantity, Relation, and Manner. The resulting coherence judgments are then aggregated and related to annotation overlap at both dialogue and turn level. This dual design is important because recent work on dialogue evaluation suggests that dialogue-level aggregation can obscure local failures that remain visible at the turn level [5], [7].

The contribution of the paper is therefore methodological as well as empirical. Methodologically, it proposes a way to relate symbolic annotation disagreement to pragmatic dialogue coherence without collapsing the two into a single notion of quality. Empirically, it tests whether lower annotation agreement with the original MultiWOZ labels corresponds to lower coherence in the underlying dialogue. Our results show that this relationship is not stable across levels of analysis or overlap metrics: at the dialogue level it remains weak, while at the turn level it becomes more differentiated and is most clearly visible for BLEU-1. The broader implication is that annotation agreement and pragmatic coherence should be treated as related but distinct evaluative signals, and that relying on only one of them risks obscuring important properties of dialogue quality.

2 RELATED WORK

This section situates the present study at the intersection of three strands of research: the annotation history of MultiWOZ, the use of large language models as annotators, and the use of large language models as dialogue evaluators. These areas are directly relevant because the paper does not only ask whether LLM-generated annotations align with an existing benchmark layer, but also whether disagreement with that layer is meaningfully related to dialogue coherence.

2.1 MultiWOZ and annotation quality

MultiWOZ remains one of the central benchmarks for task-oriented dialogue research, but its annotation history makes clear that benchmark popularity does not guarantee annotation reliability. This is especially visible in MultiWOZ 2.4, which was introduced to correct incorrect and inconsistent state annotations in the validation and test portions of MultiWOZ 2.1 while leaving the training data unchanged. Ye et al. report that the corrected states affect more than 40% of turns and more than 65% of dialogues, which shows that annotation disagreement in MultiWOZ is not a marginal issue but a structurally relevant property of the dataset itself. In this context, agreement with the original annotation layer cannot simply be equated with dialogue quality; rather, it should be treated as one evaluative dimension among others [1].

2.2 LLMs as annotators

Recent work on LLM-based annotation suggests that large language models can be useful annotation tools, but that their performance depends strongly on the task and on how agreement is defined. Pavlovic and Poesio survey twelve studies on LLM labelling and argue that the main advantages lie in speed and scalability, while the main limitations include representational mismatch, bias, prompt sensitivity, and weaker performance on more subjective tasks [3]. In a dialogue-relevant setting, Movva et al. compare GPT-4 and human annotations of conversational safety and show that even when alignment is substantial, the model can still apply systematically different standards from human annotators [2]. More recent work by Schroeder et al. likewise cautions that LLM assistance can influence human annotation behavior in subjective tasks rather than simply making it more reliable, while Calderon et al. propose a statistical framework for deciding when LLM annotations are strong enough to substitute for human ones at all [4], [8]. Taken together, this literature suggests that the relevant question is not only whether an LLM can annotate, but also what exactly its annotations align with and under which methodological conditions that alignment should count as sufficient.

2.3 LLMs as dialogue evaluators and coherence judges

Recent work on dialogue evaluation is directly relevant to our study because it supports two design decisions in our pipeline: the use of an LLM as a coherence judge and the separation of turn-level from dialogue-level analysis. Zhang et al. show that large language models can function as automatic dialogue evaluators, although their effectiveness depends on model choice and evaluation setup [6]. More importantly for the present paper, SODA-EVAL demonstrates the value of large-scale turn-level dialogue assessment and highlights that coherence problems can persist even when responses appear generally fluent and relevant [5]. TD-EVAL makes a closely related argument for task-oriented dialogue specifically: it proposes a framework that combines fine-grained turn-level analysis with dialogue-level comparison, because dialogue-level evaluation alone can miss important local failures [7]. These studies are directly relevant for our work, since they motivate both our Grice-based LLM judge and our decision to compare the relationship between annotation agreement and coherence at both dialogue and turn level.

A particularly relevant example is the work of Kazi et al., who use LLMs as user-agents for evaluating task-oriented dialogue systems in a dynamic multi-turn setting. Their framework combines an LLM-based user simulator with automatic evaluation metrics such as task completion, naturalness, coherence, and dialogue-level diversity, and it explicitly shows that prompt design affects the resulting evaluation behavior. This is closely related to the present study in that it treats LLMs not only as generators, but also as evaluative components within task-oriented dialogue. At the same time, the focus differs in an important respect: whereas Kazi et al. evaluate TOD systems through simulated interaction and multi-metric assessment, the present paper asks whether disagreement between an LLM-based re-annotation and the original MultiWOZ annotation layer can function as an indirect signal of pragmatic incoherence in the underlying dialogue [9].

3 RESEARCH QUESTIONS AND HYPOTHESES

The present study examines whether symbolic annotation agreement and pragmatic dialogue coherence capture the same underlying quality dimension in MultiWOZ. While recent work has shown both that MultiWOZ annotations are historically noisy [1] and that LLMs can be used as annotators and evaluators under certain conditions [2], [4], [6], it remains unclear whether agreement with an existing annotation layer actually corresponds to the coherence of the dialogue itself. Our project addresses this gap by combining an LLM-based re-annotation pipeline with a Grice-inspired coherence judge and comparing the resulting signals at different levels of analysis.

The study is guided by three research questions. RQ1 asks how closely the LLM-generated re-annotations align with the original MultiWOZ annotations under different overlap regimes. RQ2 asks whether lower symbolic agreement corresponds to lower coherence in the dialogue. RQ3 asks whether the answer to this question depends on the level of analysis, that is, whether the relationship differs between dialogue-level and turn-level evaluation. These questions reflect the central methodological concern of the paper: annotation agreement and discourse coherence may be related, but they are not necessarily interchangeable.

The main hypothesis is formulated in its strongest and most direct form at the turn level: H1: lower turn-level BLEU or Jaccard overlap is associated with lower turn-level coherence. A second, more methodological hypothesis follows from recent work on dialogue evaluation: H2: dialogue-level aggregation weakens or obscures the relationship between symbolic overlap and coherence by averaging over local variation within the interaction. This distinction is important because the dialogue-level and turn-level analyses in our pipeline are not merely two ways of summarizing the same result, but two different ways of operationalizing the relationship between annotation disagreement and discourse quality.

4 METHOD

This section describes the analytical pipeline used in the study. Since the paper relates symbolic annotation mismatch to pragmatic coherence, the method consists of two parallel branches: an annotation-comparison branch and a coherence-evaluation branch. The following subsections explain how the re-annotations were generated and compared, how the Grice-based coherence judgments were produced, and how both branches were aggregated and linked analytically at dialogue and turn level.

4.1 Overview of the pipeline

The overall pipeline of the study is shown in Figure 1. It consists of two parallel analytical branches that are linked only in the final evaluation step. The first branch operates on the annotation level: LLM-generated MultiWOZ-style re-annotations are aligned with the original MultiWOZ annotation layer via shared dialogue IDs and then compared using BLEU and Jaccard under strict, normalized, and lenient matching conditions. The second branch operates on the raw dialogue level: the original customer–expert exchanges are evaluated with a Grice-based, role-aware coherence judge that scores each next turn in context and produces unit-level coherence labels. These coherence annotations are then aggregated to turn and dialogue level. The two branches are finally merged either by dialogue ID or by dialogue ID and turn index in order to test whether lower symbolic annotation agreement is associated with lower coherence.

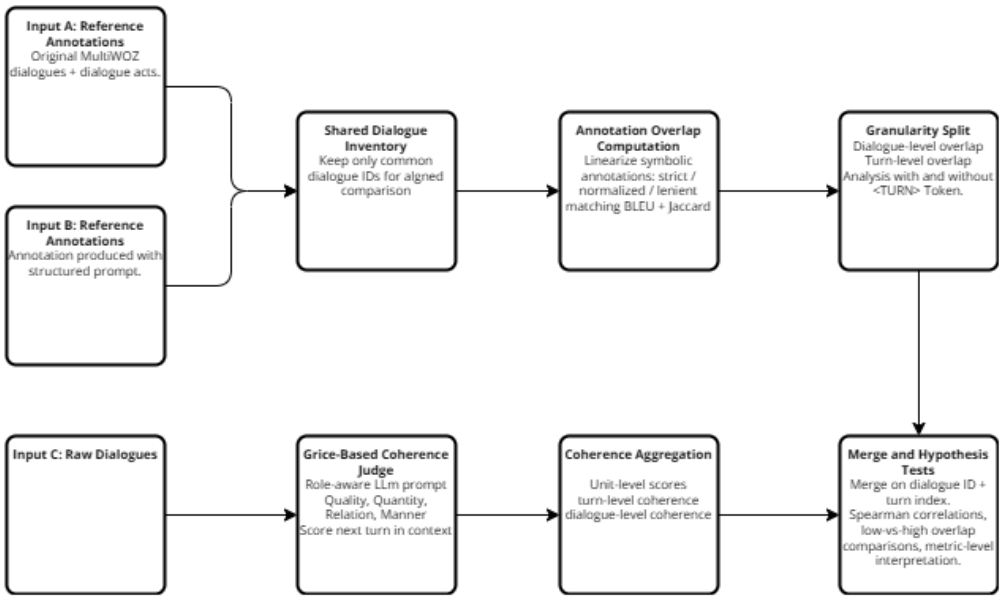

Figure 1. Overview of the analysis pipeline. The upper branch computes annotation overlap, the lower branch computes Grice-based coherence, and both branches are merged at dialogue and turn level.

4.2 LLM-based re-annotation and overlap computation

The first branch of the pipeline compares our LLM-generated MultiWOZ-style annotations with the original annotation layer of the corpus. To ensure direct comparability, we first restrict the analysis to the subset of dialogues that is available in both resources, resulting in a shared inventory of 10,438 dialogues. We then linearize the symbolic dialogue annotations into token sequences and compute overlap under three matching regimes: strict, normalized, and lenient. In the strict setting, slot values are preserved as literally as possible, so differences in casing, placeholder values, or formatting remain visible as annotation mismatches. In the normalized setting[1], values are lowercased and common null-like placeholders such as *none*, *null*, or *n/a* are collapsed, so that superficial representational variation does not count as substantive disagreement. In the lenient setting, we apply this normalization more permissively by additionally harmonizing selected boolean-like slot values, for example in fields such as *wifi*, *internet*, or *parking*, where variants such as *true*, *yes*, *1*, and in some cases *free* or *available* are

[1] The exact normalization rules for the three matching regimes are specified in the project's analysis notebook and repository implementation.

treated as equivalent. This three-way design is methodologically important because annotation disagreement in MultiWOZ can arise not only from genuine semantic mismatch, but also from inconsistent value representation, normalization choices, and incomplete or variant forms of the same underlying information.

To quantify agreement, we use Jaccard and BLEU as complementary overlap measures. Jaccard captures how much symbolic content is shared between the reference and generated annotations, independent of token order, whereas BLEU captures sequential n-gram overlap and is therefore more sensitive to how annotation content is arranged in the linearized representation. We use both measures because they emphasize different aspects of similarity: Jaccard is useful for asking whether largely the same annotation information is present at all, while BLEU is useful for asking whether that information is encoded in a similar local sequence structure. BLEU also remains a common comparison metric in dialogue-generation work; for example, Wagner and Ultes report BLEU alongside dialogue-act-based evaluation when assessing controllability in task-oriented dialogue generation [10]. In the present study, however, BLEU is not interpreted as a direct measure of dialogue quality, but as one overlap signal within a broader evaluation design that is later contrasted with Grice-based coherence.

A further methodological issue concerns the synthetic <TURN> token that was initially inserted when full dialogues were serialized into a single sequence. This marker is useful for preserving turn boundaries during dialogue-level sequence comparison, because it prevents adjacent annotations from different turns from collapsing into one uninterrupted token stream. At the same time, <TURN> is not part of the annotation content itself, but an analytical boundary marker introduced during preprocessing. In our own robustness checks, its inclusion had only a modest effect on Jaccard, but a much stronger effect on BLEU, indicating that BLEU in particular is sensitive to shared structural scaffolding in the serialized dialogue representation. For that reason, the main overlap analyses reported in this paper use the no-<TURN> version, while the boundary-marked version is retained only as a robustness check. The original corpus-level BLEU summary corresponds to the boundary-marked representation.

4.3 Grice-based coherence judge

The second branch of the pipeline evaluates the raw dialogue turns with a Grice-inspired coherence judge. Unlike the annotation-comparison branch, which operates on symbolic labels, this stage works directly on the original customer–expert exchanges and asks whether a turn is coherent in its immediate dialogue context. The judge is designed as a role-aware prompt for task-oriented dialogue: it evaluates only the next turn, distinguishes between CUSTOMER and EXPERT behavior, and operationalizes coherence through the Gricean dimensions of Quality, Quantity, Relation, and Manner. This design is central to the study, because it allows us to compare symbolic annotation agreement with a second, explicitly pragmatic notion of dialogue quality rather than relying on overlap measures alone[2].

The prompt constrains the model to a fixed seven-point rating scale consisting of Very poor, Poor, Weak, Neutral, Good, Very good, and Excellent. In addition, it allows a single turn to be split into multiple information units when an utterance contains several pragmatically distinct contributions. This is important because turns in task-oriented dialogue often combine more than one communicative function: a speaker may, for example, confirm prior information, introduce a new constraint, and ask a follow-up question within the same turn. The judge therefore returns structured JSON output at the unit level, including the relevant dialogue and turn identifiers, the speaker role, a short reasoning field, and the assigned score. This design makes the coherence annotations interpretable while also allowing them to be aggregated flexibly to turn or dialogue level depending on the experiment.

The scoring process iterates through the corpus turn by turn and constructs the dialogue context incrementally, so that each judgment is based only on the preceding interaction and the current candidate turn. In this way, the coherence judge mirrors the local interpretive situation of dialogue processing rather than evaluating the conversation retrospectively from a global summary. The resulting outputs are written in a JSONL-style format, which preserves the individual coherence judgments in a transparent and inspectable form and allows the later aggregation step to be carried out reproducibly.

4.4 Aggregation and statistical analysis

Since the raw output of the coherence judge is produced at the information-unit level, an additional aggregation step is required before it can be compared systematically with the annotation-overlap measures. We therefore map the ordinal score labels to a numeric seven-point scale ranging from Very poor to Excellent and aggregate the resulting values first to the turn level and, for the dialogue-level experiment, then further to the dialogue level. This yields two complementary coherence representations: a fine-grained turn-level measure that preserves local variation within a dialogue and a dialogue-level measure that summarizes the overall pragmatic quality of an interaction.

[2] All notebooks and results can be found at: https://github.com/ndl12ben/MMI_Algorithmen_WS25_26

The turn-level aggregation is used to test the main hypothesis in its most direct form, namely whether lower annotation overlap corresponds to lower coherence within the same turn. For this purpose, the overlap branch and the coherence branch are merged by dialogue ID and turn index. In the dialogue-level experiment, the same logic is applied at a more global level: turn-level coherence values are aggregated to dialogue-level summaries, which are then merged with dialogue-level BLEU and Jaccard scores by dialogue ID. The dialogue-level analysis therefore serves as a broader comparative perspective, whereas the turn-level analysis provides the more precise operationalization of the hypothesized relationship between annotation disagreement and coherence.

To test the relationship between overlap and coherence, we use Spearman rank correlations. This choice is motivated by the nature of the data: the coherence signal originates in an ordinal rating scale, and the relationship of interest is monotonic rather than necessarily linear. In addition to the correlation analysis, we conduct grouped comparisons in order to make the direction and practical size of the relationship easier to interpret. For these comparisons, cases are divided separately for each overlap metric and matching regime into quartiles. The lower-overlap group corresponds to the bottom quartile and the higher-overlap group to the top quartile, while the middle 50% of cases is excluded from this specific comparison. Coherence values are then compared between the two groups. These grouped analyses complement the correlation coefficients by showing more directly whether turns or dialogues with relatively weak annotation agreement also tend to receive lower coherence scores.

A final analytical decision concerns the treatment of the overlap metrics themselves. Because BLEU-1, BLEU-4, and Jaccard capture different aspects of similarity, they are not assumed to be interchangeable. Instead, they are analyzed separately and interpreted in relation to their respective formal properties. This is especially important in the present study because the central question is not simply whether agreement exists, but whether different notions of annotation overlap align with pragmatic coherence in the same way. The analysis is therefore designed not only to test for association, but also to reveal where different overlap metrics converge or diverge in their relationship to coherence.

5 RESULTS

This section presents the empirical results of the study in three steps. It begins with the degree of agreement between the LLM-generated re-annotations and the original MultiWOZ labels, then examines the relationship between overlap and coherence at the dialogue level, and finally turns to the more fine-grained turn-level analysis. This progression mirrors the structure of the pipeline and moves from descriptive overlap patterns to the central hypothesis test of the paper.

5.1 Annotation agreement with the original MultiWOZ labels

The first step of the analysis examined how closely our LLM-generated MultiWOZ-style annotations aligned with the original annotation layer of the corpus. As shown in Table 1, the same overall pattern appears at both dialogue and turn level: agreement is lowest under strict matching and higher under normalized and lenient matching. At the dialogue level, the comparison over the shared set of 10,438 dialogues yielded mean Jaccard values of 0.34 under strict, 0.38 under normalized, and 0.39 under lenient matching, with the corresponding BLEU values increasing from 20.96 to 25.86 and 26.20. This ordering indicates that a substantial portion of the disagreement between our annotations and the original MultiWOZ labels is sensitive to representational choices such as normalization and value harmonization rather than reflecting only categorical semantic mismatch.

The turn-level results show the same tendency, but also make clear that agreement remains clearly incomplete. Across approximately 143,000 matched turns per mode, mean Jaccard values rise from 0.28 under strict matching to 0.31 under normalized and 0.32 under lenient matching. Mean BLEU-1 values increase from 36.52 to 40.79 and 41.05, while mean BLEU-4 values rise from 23.02 to 26.81 and 27.14. At the same time, exact turn matches remain comparatively rare, ranging only from 2.73% under strict matching to 3.70% and 3.72% under normalized and lenient matching. Taken together, these results suggest that the re-annotation pipeline achieves substantial but clearly incomplete agreement with the original annotation layer, and that much of this agreement is local rather than exact. This makes the pipeline suitable for the next analytical step, namely testing whether symbolic disagreement with the original annotation layer corresponds to pragmatic incoherence in the dialogue itself.

Table 1: Annotation agreement between the LLM-generated re-annotations and the original MultiWOZ labels at dialogue and turn level.

Level	Mode	N	Mean Jaccard	Mean BLEU	Mean BLEU-1	Mean BLEU-4	Exact match (%)
Dialogue	Strict	10438	0.34	20.96	—	—	—
Dialogue	Normalized	10438	0.38	25.86	—	—	—
Dialogue	Lenient	10438	0.39	26.20	—	—	—
Turn	Strict	143000	0.28	—	36.52	23.02	2.73
Turn	Normalized	143000	0.31	—	40.79	26.81	3.70
Turn	Lenient	143000	0.32	—	41.05	27.14	3.72

5.2 Dialogue-level relationship between overlap and coherence

The dialogue-level analysis tested whether dialogues with lower annotation agreement also tend to exhibit lower average coherence. For this purpose, dialogue-level overlap scores were merged with dialogue-level coherence summaries across all 10,438 dialogues, resulting in 31,314 merged rows across the three matching regimes. The relationship was then examined with Spearman rank correlations between dialogue-level overlap measures and the mean Grice-based coherence score per dialogue.

Overall, the dialogue-level results do not support a meaningful positive relationship between annotation agreement and coherence. As shown in Figure 2a, both overlap measures remain weak across all three matching regimes and do not display the expected positive pattern. For Jaccard, the correlations with dialogue-level mean coherence were consistently negative, but very small in size: $\rho = -0.098$ under strict**,** $\rho = -0.079$ under normalized, and $\rho = -0.065$ under lenient matching. Although these effects reached statistical significance due to the large number of dialogues, their magnitude is too small to indicate a substantively strong association. For BLEU, the relationship was even weaker: the corresponding correlations were $\rho = -0.010$ for strict**,** $\rho = -0.023$ for normalized, and $\rho = -0.013$ for lenient matching, with only the normalized condition reaching statistical significance and even there remaining close to zero.

Taken together, these findings suggest that dialogue-level aggregation obscures any simple relationship between symbolic annotation overlap and pragmatic coherence. Dialogues whose annotations differ more strongly from the original MultiWOZ layer do not, on average, appear substantially less coherent according to the Grice-based judge. This dialogue-level result is therefore best interpreted as a negative or null finding, and it motivates the more fine-grained turn-level analysis reported next.

5.3 Turn-level relationship between overlap and coherence

The turn-level analysis provides a more differentiated picture. After merging turn-level overlap scores with turn-level coherence summaries, the resulting dataset comprised 428,562 merged rows, corresponding to 142,854 unique turns across the shared dialogue set. In contrast to the dialogue-level analysis, the turn-level results show that the relationship between annotation agreement and coherence depends strongly on the overlap metric used. This metric-sensitive contrast is visible in Figure 2b.

For Jaccard, the correlations with turn-level mean coherence were again negative, although weak in size: $\rho = -0.098$ under strict, $\rho = -0.051$ under normalized, and $\rho = -0.050$ under lenient matching. Thus, higher set-based symbolic overlap was not associated with higher turn-level coherence in the expected direction. For BLEU-1, however, the pattern was clearly different: the correlations were consistently positive, with $\rho = 0.167$ for strict, $\rho = 0.143$ for normalized, and $\rho = 0.142$ for lenient matching. Although these effects remain modest in magnitude, they provide the clearest support in the study for the idea that stronger local annotation similarity can coincide with higher local coherence. BLEU-4 was less stable: under strict matching the correlation remained weakly positive ($\rho = 0.052$), whereas under normalized and lenient matching it became slightly negative ($\rho = -0.016$ and $\rho = -0.015$, respectively). Taken together, these results indicate that H1 is only partially supported: evidence for the expected positive relationship emerges only for turn-level BLEU-1, whereas Jaccard shows the opposite pattern and BLEU-4 remains weak and unstable across matching regimes.

This metric sensitivity also becomes visible in the grouped comparisons between low-overlap and high-overlap cases shown in Figure 2c. Here, lower-overlap and higher-overlap refer to the bottom and top quartile of the respective overlap metric within each matching regime. When turns were divided into these groups, Jaccard again showed the opposite of the expected direction: higher-overlap turns had slightly lower coherence on average, with differences of -0.091 under strict, -0.078 under normalized, and -0.079 under lenient matching. BLEU-1, by contrast, showed the clearest positive pattern in the study: higher-overlap turns received clearly higher coherence scores, with mean differences of +0.301, +0.288, and +0.282 across the three matching regimes. BLEU-4 showed a weaker and less stable version of the same tendency, with differences of +0.217 under strict, +0.063 under normalized, and +0.067 under lenient matching. These grouped results reinforce the correlation analysis by showing that local symbolic similarity and local pragmatic coherence align only to a limited extent, and primarily when overlap is measured through a relatively surface-sensitive local metric such as BLEU-1 rather than through set-based overlap or higher-order n-gram precision.

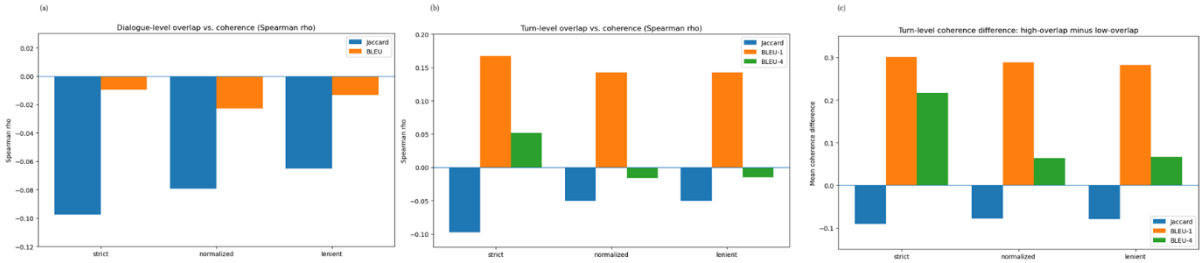

Figure 2. Relationship between annotation overlap and Grice-based coherence across analysis levels and metrics. (a) Dialogue-level Spearman correlations remain weak and do not show a meaningful positive relationship between overlap and coherence. (b) Turn-level correlations reveal a more differentiated pattern: BLEU-1 shows the clearest positive relationship with coherence, while Jaccard remains weakly negative and BLEU-4 small and unstable. (c) Mean coherence differences between high-overlap and low-overlap turns reinforce this metric-sensitive turn-level pattern.

6 DISCUSSION

The results show that annotation mismatch is not a simple or stable proxy for dialogue incoherence. At the same time, they reveal patterns that are methodologically informative, especially with regard to analytical granularity and metric choice. This section interprets the findings in four steps: first by reconsidering annotation mismatch as a possible signal of incoherence, then by explaining why dialogue-level aggregation weakens the pattern, next by examining the role of the overlap metric itself, and finally by drawing broader methodological implications for LLM-based annotation and dialogue evaluation.

6.1 Annotation mismatch as a possible signal of incoherence

The central question of this study is not whether annotation overlap itself constitutes coherence, but whether annotation mismatch can function as an indirect signal of underlying dialogue difficulty. More specifically, we ask whether dialogues or turns that are pragmatically noisier, more confusing, or less cooperative in context are also more likely to produce divergence between our LLM-generated re-annotations and the original MultiWOZ annotation layer. The guiding assumption is therefore diagnostic: if a dialogue turn is locally clear and interactionally well-formed, it should be easier to annotate consistently, whereas pragmatically weak or ambiguous turns may invite greater disagreement between two annotation versions.

Our results support this assumption only to a limited extent. At the dialogue level, lower overlap between our annotations and the original MultiWOZ labels does not meaningfully correspond to lower coherence. The observed correlations remain weak and do not show the expected positive pattern. At the turn level, however, the picture is more differentiated: turns with higher BLEU-1 overlap also tend to receive higher coherence scores, which means that locally clearer turns appear somewhat easier to annotate consistently. At the same time, this effect is only modest in size and does not generalize across all overlap measures, since Jaccard and BLEU-4 do not show the same stable positive relationship. Annotation mismatch can therefore not be treated as a uniform proxy for pragmatic incoherence.

This finding matters because it shows that symbolic agreement with an annotation layer does not automatically track pragmatic quality in context. This matters particularly in MultiWOZ, whose annotation history includes substantial corrections and known inconsistencies [1].

6.2 Why dialogue-level aggregation weakens the pattern

A second important finding concerns the level of analysis. At the dialogue level, the relationship between annotation mismatch and coherence is weak or effectively absent, whereas the turn-level analysis reveals a more differentiated pattern. This suggests that the assumed relationship is too local to remain clearly visible once both overlap and coherence are averaged over complete dialogues. A dialogue may contain several pragmatically unclear, contradictory, or weak turns without this lowering its overall average coherence enough to produce a strong dialogue-level signal. Conversely, a dialogue can contain annotation mismatches that are locally meaningful without these mismatches accumulating into a clearly incoherent interaction when viewed as a whole. In this sense, the dialogue-level result should not simply be read as a failed hypothesis test, but as evidence that the relevant relationship is diluted by aggregation.

This interpretation also follows from the design of the coherence branch itself. The Grice-based judge evaluates the next turn only in its immediate dialogue context and allows a turn to be split into multiple information units where necessary. The scoring procedure is therefore explicitly oriented toward local discourse interpretation rather than toward retrospective global summarization. It is therefore methodologically plausible that the relationship between coherence and annotation mismatch becomes more visible at the turn level, where both variables still refer to the same local interactional unit, than at the dialogue level, where both have already been smoothed into broader summary values.

The dialogue-level analysis nevertheless remains useful. It shows that annotation disagreement does not automatically scale up into a broader signal of dialogue incoherence. This is an important negative finding, because it places a limit on how far overlap-based disagreement can be interpreted as an indicator of underlying dialogue difficulty. If annotation mismatch is to be used diagnostically at all, the present results suggest that it is more informative as a local than as a global signal.

6.3 The overlap metric matters

A third important result of the study is that the relationship between annotation mismatch and coherence depends strongly on how overlap is measured. The turn-level analysis shows that BLEU-1, BLEU-4, and Jaccard do not behave in the same way, even though all three are intended to capture some form of similarity between our re-annotations and the original MultiWOZ labels. This means that the paper is not simply comparing “overlap” to “coherence” in general, but rather examining whether different formalizations of overlap are differently related to pragmatic dialogue quality.

The clearest positive pattern emerges for BLEU-1. At the turn level, turns with higher BLEU-1 overlap also tend to receive higher coherence scores, and the grouped comparisons show the same tendency: higher-overlap turns have noticeably higher average coherence than lower-overlap turns. This suggests that BLEU-1 captures a type of local symbolic similarity that aligns, at least to some extent, with pragmatic clarity. One plausible interpretation is that turns that are easier to understand and annotate consistently also produce annotation sequences that remain locally closer to the original reference. In this sense, BLEU-1 appears to function as a weak but meaningful indicator of local interactional straightforwardness.

By contrast, Jaccard does not show the expected pattern. Instead of a positive relationship, the results are weakly negative both in the correlation analysis and in the grouped comparisons. This indicates that simple set-based symbolic overlap is not a reliable proxy for coherence in the present setting. A likely reason is that Jaccard ignores token order and therefore captures only whether similar annotation content is present at all, not whether it is structured in a locally similar way. For short task-oriented dialogue turns, however, local sequencing and immediate formulation may matter more than general content overlap. BLEU-4 occupies an intermediate position: it is more sensitive to sequence structure than Jaccard, but it appears too strict and unstable for many short turns, where higher-order n-gram matching quickly becomes sparse.

These differences have an important methodological consequence. The results suggest that overlap metrics should not be treated as interchangeable signals of annotation quality or underlying dialogue difficulty. In the present study, BLEU-1 is the only metric that shows a reasonably consistent positive relationship with turn-level coherence, whereas Jaccard and BLEU-4 do not. The broader implication is that whether annotation mismatch appears informative depends not only on the data and the hypothesis, but also on the formal properties of the metric used to operationalize similarity in the first place.

6.4 Methodological implications

Taken together, the findings suggest that symbolic overlap and coherence are best evaluated separately and only then related analytically. Agreement with an existing annotation layer captures how closely a generated symbolic representation reproduces a corpus-internal label structure. Coherence, by contrast, captures whether a turn is pragmatically appropriate in context, that is, whether it is relevant, sufficiently informative, non-contradictory, and interactionally well-formed. The present results show that one cannot simply be inferred from the other. In particular, annotation mismatch does not function as a stable general-purpose indicator of dialogue incoherence, but at best as a partial and metric-dependent local signal.

This has methodological consequences for work on LLM-based annotation as well as for dialogue evaluation. If overlap with a reference annotation layer is used as the only success criterion, then a model may appear to perform well without necessarily capturing the pragmatic properties of the underlying interaction. Conversely, disagreement with a reference layer should not automatically be interpreted as evidence that the dialogue itself is poor or incoherent. In this respect, the present study supports a more differentiated evaluation design in which symbolic agreement and discourse-level coherence are examined separately and then related analytically, rather than being collapsed into a single notion of quality.

A further implication concerns analytical granularity. The comparison between dialogue-level and turn-level results indicates that local interactional phenomena can easily disappear under broader aggregation. For future work on dialogue annotation and evaluation, this suggests that turn-level analysis should not be treated merely as an optional refinement, but as a necessary level of observation whenever the aim is to connect annotation behavior with discourse properties. More generally, the study shows that evaluation design benefits from preserving intermediate representations and multiple analysis levels, since these make it possible to identify where different quality signals converge and where they diverge.

The discussion has shown that annotation mismatch can, under some conditions, function as a local signal of dialogue difficulty, but not as a stable or general proxy for incoherence. This interpretation, however, depends on a number of methodological decisions and project-specific constraints that should be made explicit before broader conclusions are drawn.

7 LIMITATIONS AND FUTURE WORK

Despite the methodological value of the current pipeline, several limitations must be acknowledged. These concern the robustness of the coherence judge, the construction of the prompts, the quality of the reference annotation layer, and the overall scope of the project. At the same time, these limitations also point directly toward the most important avenues for future work, especially the validation of the coherence judge and the development of a more formal rule-based comparison framework.

7.1 Limitations of the coherence judge

A first limitation concerns the robustness of the coherence judge itself. The Grice-based scoring procedure was implemented in a single-model setup and was not systematically repeated across multiple runs or multiple judge models. The coherence scores reported in this paper should therefore be understood as one model-dependent operationalization of pragmatic quality rather than as a fully stabilized estimate. This matters because LLM-based judgments can be sensitive to model choice, prompting, and generation variability. Future work should therefore test the reliability of the coherence scores across repeated runs, alternative model backends, and, ideally, comparison with human judgments on a manually inspected subset.

A related limitation is that the present study does not include an independent human gold standard for coherence. Although the Grice-based prompt was designed to capture relevant discourse properties in a structured way, the current analysis does not yet establish how closely the resulting LLM scores align with human assessments of the same turns. A next step should therefore be the manual evaluation of a smaller sample of turns and dialogues in order to validate the extent to which the judge captures coherence in a way that is meaningful beyond the model's own internal criteria.

7.2 Limitations of prompt construction

A second limitation concerns prompt design. Both the annotation prompts and the Grice-based judging prompts were developed iteratively through exploratory trial and error until the resulting outputs appeared qualitatively plausible and useful for the purposes of the study. This was a practical and productive strategy within the seminar context, but it also means that the final prompts were not derived from a formally validated annotation protocol. As a result, the study necessarily contains a researcher-designed interpretive layer at the level of prompt formulation. Future work should therefore examine prompt sensitivity more systematically, compare alternative prompt formulations, and evaluate whether the present results remain stable under controlled prompt variation.

7.3 Limitations of the reference annotation layer

A further limitation concerns the reference annotation layer against which our re-annotations are compared. The overlap analyses use the original MultiWOZ annotation layer as their point of comparison, even though the dataset is known to contain annotation inconsistencies and has undergone substantial correction in later versions. MultiWOZ 2.4 was introduced precisely because earlier versions still contained incorrect and inconsistent state annotations, including context mismatch, missing annotations, and incomplete or wrongly represented values [1]. This means that disagreement with the reference annotation cannot automatically be interpreted as annotation failure on the side of the LLM-generated labels. In at least some cases, divergence may reflect ambiguity or weakness in the corpus annotation itself rather than reduced quality in the re-annotation

pipeline. For future work, it would therefore be valuable to compare against revised MultiWOZ variants or to include targeted manual inspection of disagreement cases.

7.4 Project scope and future work

Finally, the scope of the project itself imposes an important limitation. The present paper emerged as a seminar-based subproject within a broader research idea that originally aimed to develop a more formalized rule-based coherence framework and to compare that framework systematically with the open LLM-based judge used here. Due to time, staffing, and participant constraints, this broader design could not be completed within the seminar setting. The present study should therefore be understood as a focused proof-of-concept rather than as the full realization of the larger research program.

This limitation also points directly toward future work. The most important next step would be to formalize coherence criteria in a more explicit rule-based scheme and to compare that structured approach with the open LLM-based Grice judge used in the present study. Such a comparison would make it possible to test not only whether LLM-based coherence judgments are useful, but also whether they are more or less sensitive than a formal rule set to local dialogue difficulty. In addition, future work should extend the present analysis by validating the judge against human ratings, comparing multiple models and multiple runs, and further examining disagreement cases where annotation mismatch and coherence clearly diverge.

8 CONCLUSION

This paper examined whether annotation mismatch can function as an indirect signal of dialogue incoherence in MultiWOZ. To do so, it combined an LLM-based re-annotation pipeline with a Grice-inspired coherence judge and related both signals at dialogue and turn level. The results show that annotation mismatch is not a stable general proxy for incoherence: at the dialogue level, the relationship remains weak, while at the turn level it appears only partially and depends strongly on the overlap metric used. The main methodological implication is that overlap with an existing annotation layer does not on its own capture discourse quality in context. Future work should therefore validate coherence judgments more systematically and compare the present LLM-based approach with a more formal rule-based coherence framework.

9 REFERENCES:

[1] F. Ye, J. Manotumruksa, and E. Yilmaz, "MultiWOZ 2.4: A Multi-Domain Task-Oriented Dialogue Dataset with Essential Annotation Corrections to Improve State Tracking Evaluation," in *Proceedings of the 23rd Annual Meeting of the Special Interest Group on Discourse and Dialogue*, O. Lemon, D. Hakkani-Tur, J. J. Li, A. Ashrafzadeh, D. H. Garcia, M. Alikhani, D. Vandyke, and O. Dušek, Eds., Edinburgh, UK: Association for Computational Linguistics, Sep. 2022, pp. 351–360. doi: 10.18653/v1/2022.sigdial-1.34.

[2] R. Movva, P. W. Koh, and E. Pierson, "Annotation alignment: Comparing LLM and human annotations of conversational safety," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Y. Al-Onaizan, M. Bansal, and Y.-N. Chen, Eds., Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 9048–9062. doi: 10.18653/v1/2024.emnlp-main.511.

[3] M. Pavlovic and M. Poesio, "The Effectiveness of LLMs as Annotators: A Comparative Overview and Empirical Analysis of Direct Representation," in *Proceedings of the 3rd Workshop on Perspectivist Approaches to NLP (NLPerspectives) @ LREC-COLING 2024*, G. Abercrombie, V. Basile, D. Bernadi, S. Dudy, S. Frenda, L. Havens, and S. Tonelli, Eds., Torino, Italia: ELRA and ICCL, May 2024, pp. 100–110. Accessed: Mar. 23, 2026. [Online]. Available: https://aclanthology.org/2024.nlperspectives-1.11/

[4] H. Schroeder, D. Roy, and J. Kabbara, "Just Put a Human in the Loop? Investigating LLM-Assisted Annotation for Subjective Tasks," in *Findings of the Association for Computational Linguistics: ACL 2025*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds., Vienna, Austria: Association for Computational Linguistics, Jul. 2025, pp. 25771–25795. doi: 10.18653/v1/2025.findings-acl.1323.

[5] J. Mendonça, I. Trancoso, and A. Lavie, "Soda-Eval: Open-Domain Dialogue Evaluation in the age of LLMs," Oct. 04, 2024, *arXiv*: arXiv:2408.10902. doi: 10.48550/arXiv.2408.10902.

[6] C. Zhang, L. F. D'Haro, Y. Chen, M. Zhang, and H. Li, "A comprehensive analysis of the effectiveness of large language models as automatic dialogue evaluators," in *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence*, in AAAI'24/IAAI'24/EAAI'24, vol. 38. AAAI Press, Feb. 2024, pp. 19515–19524. doi: 10.1609/aaai.v38i17.29923.

[7] E. C. Acikgoz *et al.*, "TD-EVAL: Revisiting Task-Oriented Dialogue Evaluation by Combining Turn-Level Precision with Dialogue-Level Comparisons," in *Proceedings of the 26th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, F. Béchet, F. Lefèvre, N. Asher, S. Kim, and T. Merlin, Eds., Avignon, France: Association for Computational Linguistics, Aug. 2025, pp. 113–132. Accessed: Mar. 23, 2026. [Online]. Available: https://aclanthology.org/2025.sigdial-1.7/

[8] N. Calderon, R. Reichart, and R. Dror, "The Alternative Annotator Test for LLM-as-a-Judge: How to Statistically Justify Replacing Human Annotators with LLMs," Aug. 08, 2025, *arXiv*: arXiv:2501.10970. doi: 10.48550/arXiv.2501.10970.

[9] T. Kazi, R. Lyu, S. Zhou, D. Hakkani-Tur, and G. Tur, "Large Language Models as User-Agents for Evaluating Task-Oriented-Dialogue Systems," Nov. 15, 2024, *arXiv*: arXiv:2411.09972. doi: 10.48550/arXiv.2411.09972.

[10] N. Wagner and S. Ultes, "On the Controllability of Large Language Models for Dialogue Interaction," in *Proceedings of the 25th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, T. Kawahara, V. Demberg, S. Ultes, K. Inoue, S. Mehri, D. Howcroft, and K. Komatani, Eds., Kyoto, Japan: Association for Computational Linguistics, Sep. 2024, pp. 216–221. doi: 10.18653/v1/2024.sigdial-1.19.